Quantization

Ingyu Bahng

June 2026

Contents

1 Light **2**
1.1 Waves . . . 2
1.2 Photons . . . 4

2 Electron Energy **5**
2.1 Wave-Particle Duality of Electrons . . . 5
2.2 Electron Orbital Basics . . . 6
2.3 Electron Energy Emissions & Absorptions . . . 13

3 Orbital Bonding **15**
3.1 Sigma/Pi & Bonding/Antibonding Orbitals . . . 15
3.2 Diatomic Orbital Bonding . . . 19
3.3 Orbital Hybridization . . . 25
3.4 Delocalization, Resonance, and Light . . . 27

1 Light

1.1 Waves

Core Concept 1.1: Light

Light consists of oscillations of electric and magnetic fields—or **electromagnatic waves**—that travel together at the speed of light ($c \approx 3.0 \times 10^8\ m\ s^{-1}$)

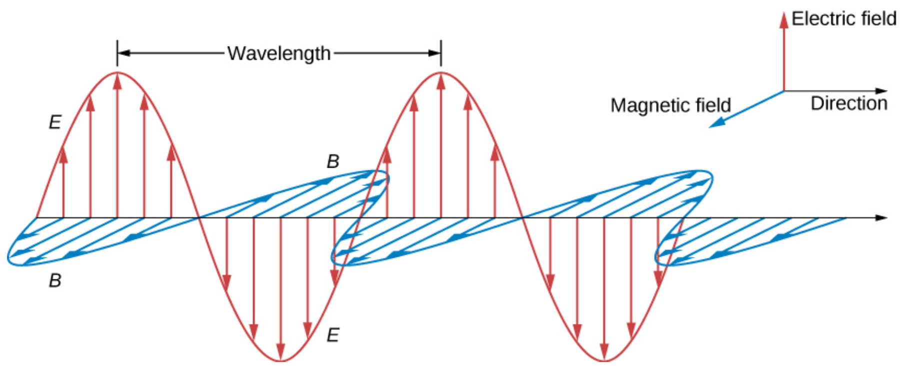

Figure 1: The respective electric and magnetic field oscillations of an electromagnetic wave

Core Concept 1.2: Frequency, Wavelength, and the Electromagnetic Spectrum

While **frequency (ν)** quantifies the number of oscillations that occur per second, **wavelength (λ)** represents the physical distance spanned by a single cycle. Thus,

$$c = \lambda \nu$$

The electromagnetic spectrum is the complete range of all electromagnetic radiation, organized by frequency or wavelength.

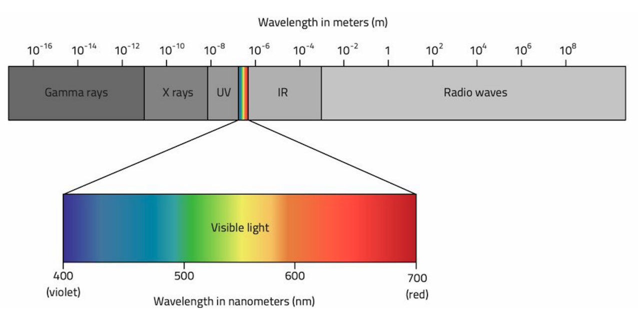

Figure 2: The electromagnetic spectrum (*top*) and the visible specturm (*bottom*)

It describes every form of light in the universe, from the vast radio waves used for broadcasting to the tiny, high-energy gamma rays emitted by nuclear reactions.

The **visible spectrum** is the portion of the EM spectrum that corresponds to EM waves with wavelengths between 400nm and 700nm.

Core Concept 1.3: Wave Power & Energy

A light wave's power and energy are two distinct characteristics.

(a) **Power** is determined by the **amplitude** of the wave and represents the rate at which the light's energy is being delivered or transferred.

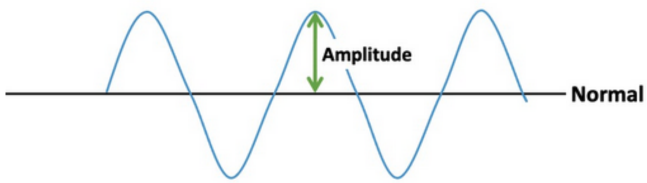

Figure 3: Wave amplitude

To understand why, it's helpful to understand the additive properties of waves or **superposition**. Waves do not "collide" and bounce off each other like billiard balls. Instead, they pass through one another, and their amplitudes add together algebraically at every point in space and time.

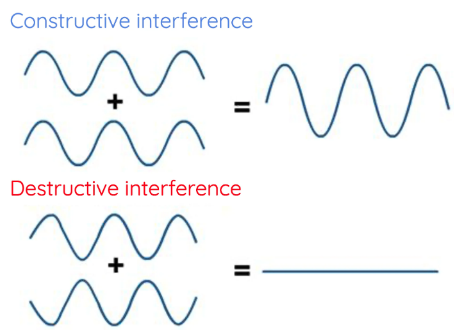

Figure 4: Constructive and destructive interference as a result of wave superposition

This means when we see an amplitude increase, it's not a single wave getting stronger, but rather many waves of the same frequency being delivered simultaneously, thus increasing the rate at which energy is delivered (*i.e., power*).

(b) **Energy** is the total amount of work a light wave can perform and is determined by the frequency of the wave. Higher frequency (*more frequent oscillations*) means higher energy, and lower frequency (*less frequent oscillations*) means lower energy. Thus

$$E_{gamma} > E_{Xrays} > E_{ultraviolet}$$

To understand the distinction between wave power and energy, imagine a stream of marbles moving through space at constant speed. Increasing each marble's mass will increase its kinetic energy on impact, thus increasing the work it can perform when colliding with another system. The marble's mass is analogous to the frequency of EM waves. Now imagine adding a second stream so the rate of marble delivery doubles. This is analogous to the power of EM waves.

1.2 Photons

Core Concept 1.4: Photons

Light energy is quantized into "packets" called **photons**, meaning that the energy of light is not a continuous, smooth flow, but is instead delivered in **discrete, indivisible "packets"**—*just as a stream of water may look continuous on the macroscale but consists of discrete, indivisible H_2O molecules on the microscale.* The energy of an individual photon is determined by the frequency of the electromagnetic wave it is part of (*as discussed above in the context of the general wave*).

$$E_{photon} = \frac{hc}{\lambda} = h\nu$$

where $h = 6.626 \times 10^{-34}\ J\,s$ is **Planck's constant**. It represents a conversion factor that relates a photon's frequency to its energy.

Core Concept 1.5: Photoelectric Effect

Prior to the discovery of photons, the leading theory was the **Classical Wave Theory**, which viewed light as a continuous wave carrying energy proportional to its intensity (*brightness*). This assumed that if you shine light on an atom, the e^-s should eventually "soak up" enough energy to break free regardless of the wavelength of the light.

The existence of photons was proven via the **photoelectric effect**, where the kinetic energy of an e^- begins to rise only after a certain threshold frequency, revealing that light interacts with matter in discrete, one-to-one events.

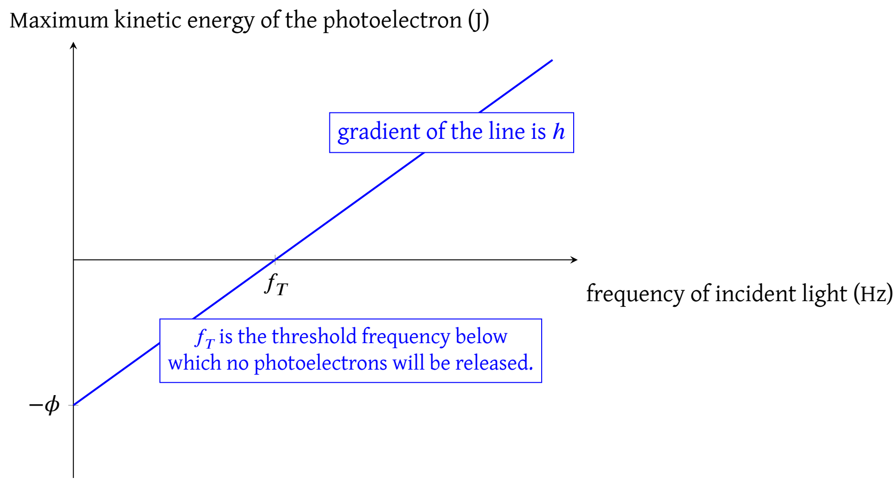

Figure 5: Photoelectric effect; maximum kinetic energy of photoelectron vs frequency of incident light

This gave rise to the **"all-or-nothing" rule**, which states that an e^- can only absorb one photon at a time and thus cannot accumulate energy from multiple low-energy photons.

As shown in the graph above, the point where the line intersects the x-axis represents the threshold frequency at which an e^- will be ejected from the atom. Anything below this frequency will have no effect, whereas anything above will cause the e^- to be ejected with greater kinetic energy.

Core Concept 1.6: Calculating Photon Energy

Let's calculate the energy of a photon of blue light ($450nm$).

$$E_{photon} = \frac{(6.626 \times 10^{-34}\ J\,s)(3.0 \times 10^{8}\ m\,s^{-1})}{450nm} = 4.417 \times 10^{-19}\ J$$

However, instead of joules, the energy of a photon can also be measured in **eV** (electron volts), a much smaller and more convenient unit than the joule for describing energy levels of individual particles like e^-s and p^+s. It represents the amount of energy gained or lost by a single e^- as it accelerates through an electric potential difference of 1 V in a vacuum.

$$eV = 1.602 \times 10^{-19}\ J$$

Thus the energy of a photon of blue light ($450nm$) would be

$$E_{photon} = \frac{4.417 \times 10^{-19}\ J}{1.602 \times 10^{-19}\ J\,eV^{-1}} = 2.757\,eV$$

2 Electron Energy

2.1 Wave-Particle Duality of Electrons

So far in our study of general chemistry, we have exclusively treated electrons as discrete particles. This is because, for the purposes of "chemical bookkeeping," the particle model is incredibly efficient (*such as in e^- transfer within redox reactions*). However, as you move deeper into physical chemistry, the model shifts.

Here we also introduce the **wave-like nature of electrons**. Physicists have proven this wave-like state, namely through the double-slit experiment.

Core Concept 2.1: Double Slit Experiment

The **double-slit experiment** was a setup where a single e^- was shot at a double slit in a wall and would leave a mark indicating where it hit on the screen on the other side of the slits.

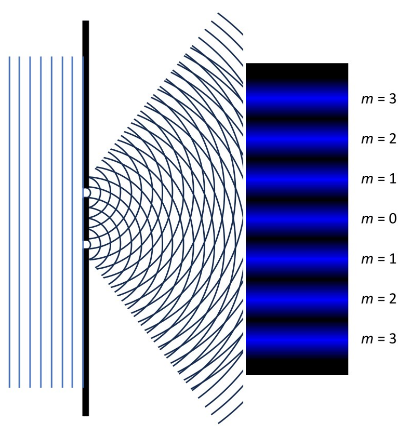

Figure 6: Double slit experiment wave interference pattern

The results of the experiment produced a diffraction pattern that could only be explained by assuming that an e^- behaves like a wave. The regions where marks were left represent regions hit by constructively interfering portions of the waves that originated from the slits. The regions with no marks represent regions hit by destructively interfering portions of the waves that originated from the slits.

If e^-s only behaved like particles, as physicists had assumed prior to this experiment, each would have passed through only one slit or the other at any given time. Over time, as you fire thousands of e^-s, you would see only two distinct clusters or "piles" of hits on the screen, directly behind each slit.

Core Concept 2.2: Single Slit Experiment

We see a similar diffraction pattern in the **single-slit experiment** as well.

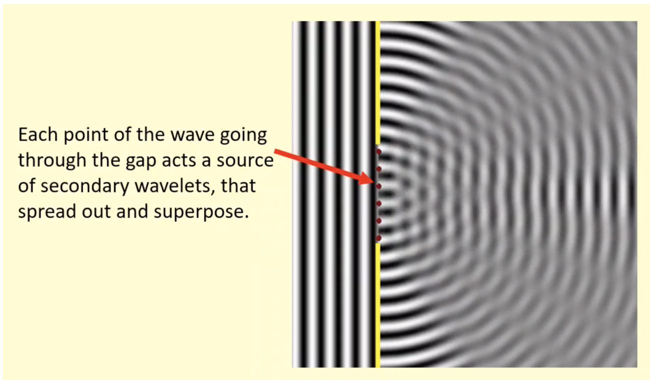

Figure 7: Single slit experiment diffraction pattern

The reason we see a diffraction pattern is the **Huygens-Fresnel principle**, which states that we don't treat a slit as just a "hole" that light passes through. Instead, every point along the width of the slit acts as a **secondary source** of spherical wavelets. These wavelets spread out in all directions from their respective points of origin. Thus the diffraction pattern we see on the screen is the interference pattern created by the superposition of all these secondary wavelets.

2.2 Electron Orbital Basics

Core Concept 2.3: Standing Waves

A **standing wave** is a wave pattern that oscillates in time but does not move in space (*confined to a restricted region*), formed by the interference of two waves of the **same frequency and amplitude** traveling **in opposite directions**.

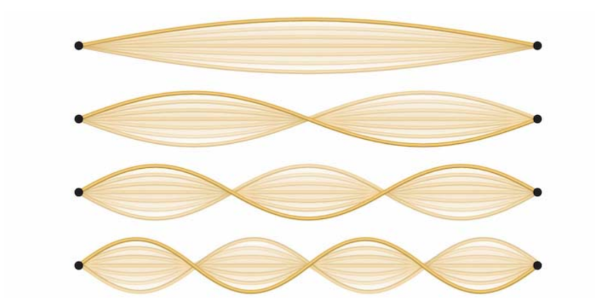

Figure 8: One-dimensional standing wave harmonics

Recognize that there are points along the standing waves shown above where the medium remains completely still and experiences zero amplitude and displacement due to destructive interference. These points are called **nodes** of the wave. Now let's take a look at a 2D standing wave.

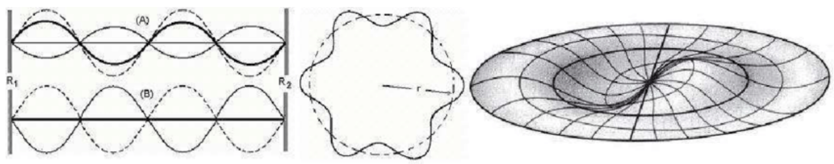

Figure 9: Two-dimensional standing wave pattern

It's harder to visualize, but just like the 1D standing wave displayed above, 2D standing waves also have nodes where the wave function sums to zero. In 2D, these nodes can be either radial or angular, as shown below.

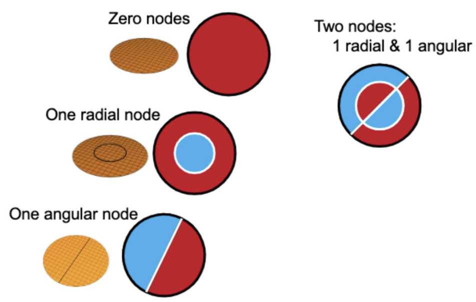

Figure 10: Radial and angular nodes of two-dimensional standing waves

It's also important to keep in mind that the energy of these standing waves increases as the node count increases because the frequency of the standing wave increases.

Core Concept 2.4: Electron Orbitals

Now we introduce **electron orbitals**, which are **3-dimensional standing waves** that represent the **likelihood** of where e^-s are located in space in the context of their wave-like characteristics rather than as particles.

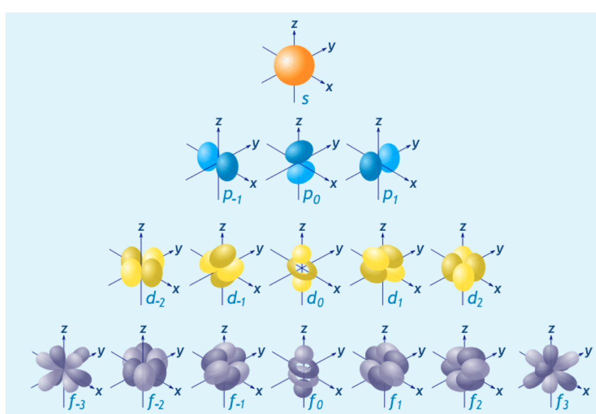

Figure 11: Shapes of s, p, d, and f orbitals

Just as a standing wave exists only within a confined region, electron orbitals are also **confined in a space determined by the Coulomb attractions** between them and the nucleus of the atom.

Core Concept 2.5: Quantum State of Electrons

Electron orbitals are categorized by their energy levels, identified with a **principal quantum number (n)** and an **angular quantum number (l)**. Electron orbitals within the same energy level are further distinguished by their **magnetic quantum number (m_l)**. Each electron within each orbital is again distinguished by its **spin quantum number (m_s)**.

(a) The **principal quantum number** n defines the electron's main energy level (shell), its average distance from the nucleus, and its size. The value $n - 1$ also represents the **total number of nodes** within the electron orbital. Generally, the higher n is, the higher the energy level of the orbital; however, there are some nuances to this based on the angular quantum number.

(b) The **angular quantum number** l defines the subshell of the electron orbital and, more importantly, the shape of the orbital by representing the **number of angular nodes**. We use letters s, p, d, f corresponding to 0, 1, 2, 3, respectively, to represent the angular quantum numbers.

Here is a schematic showing the different electron orbitals with their principal and angular quantum numbers. The arrow indicates the order from lowest (1s) to highest (7p) orbital energy levels.

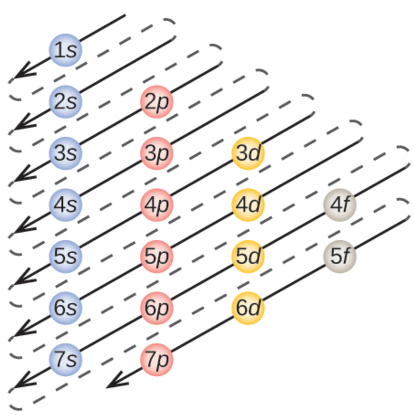

Figure 12: Electron orbital energy level ordering schematic

Let's look at an example of the 3p orbital and its corresponding principal quantum number (3) and angular quantum number (1 or p).

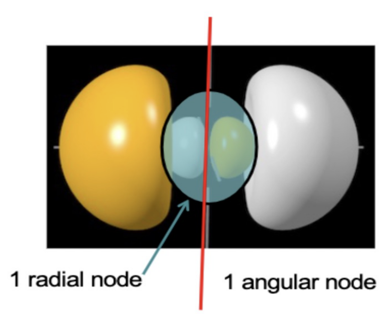

Figure 13: 3p electron orbital

As you can see, there are a total of $n - 1 = 3 - 1 = 2$ nodes, of which there are $l = 1$ angular nodes.

(c) The **magnetic quantum number** m_l simply tells us the spatial orientation of the electron orbital. The values for m_l, however, depend on the angular quantum number l. m_l can only be integers from $-l$ to l, inclusive.

l	Possible m_l
0 (s)	0
1 (p)	$-1, 0, +1$
2 (d)	$-2, -1, 0, +1, +2$
3 (f)	$-3, -2, -1, 0, +1, +2, +3$

Table 1: Magnetic quantum numbers (m_l) for each angular quantum number (l)

We revisit this figure, which illustrates the various orientations across the four angular quantum numbers.

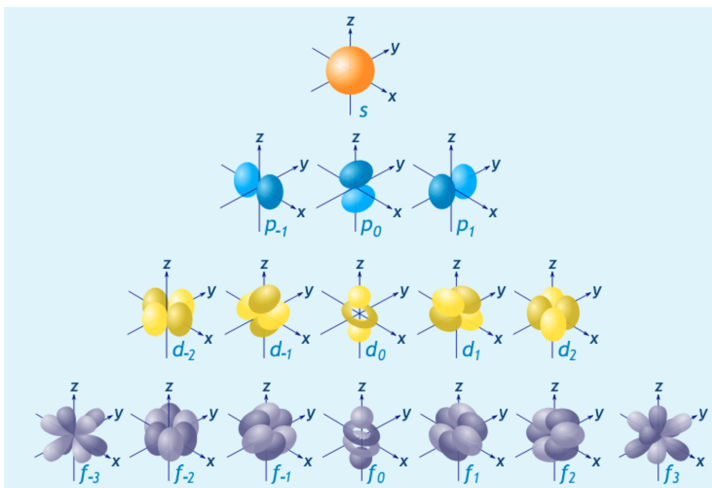

Figure 14: Shapes of s, p, d, and f orbitals

As shown, s and p orbitals possess relatively straightforward orientations—either none (*for the s orbital*) or along the x, y, or z axis (*for the p orbitals*). However, the d and f orbitals require additional orientations beyond these simple Cartesian axes. The nature of these additional orientations will not be examined in these notes.

It is sufficient to understand that the orientations of each electron orbital (*with unique principal, angular, and magnetic quantum number combinations*) formed within an atom are directed such that they remain **orthogonal** to one another. While **orthogonality** in mathematics denotes independence or perpendicularity, in the context of electron orbitals, it signifies that each orbital is **statistically non-overlapping** with all others, despite potentially occupying the same general spatial region. Consequently, the e^-s within the $2p_{-1}$ orbital will not occupy the $2p_0$ or $2p_{+1}$ orbitals, nor any other orbitals with differing angular or principal quantum numbers.

(d) Finally, we address the **spin quantum number**. The spin quantum number describes the orientation of the electron's spin within the orbital. It should be noted that nothing physically "spins" in the classical sense; rather, spin represents the orientation of the electron's intrinsic magnetic moment. This magnetic moment is quantized such that measurement of its magnitude along any given direction yields only two possible values: spin-up (+1/2) or spin-down (−1/2), which are orthogonal to each other.

Core Concept 2.6: Pauli Exclusion Principle

Each e^- within an atom possesses a unique set of quantum numbers (n, l, m_l, and m_s) that fully characterize the electron's spatial and angular orientation, as well as its spin state. This is formalized by the **Pauli Exclusion Principle**, which states that **no two e^-s can occupy the same orbital with the same spin orientation**.

Core Concept 2.7: Orbital Energy Discrepancies & Hydrogenic Atomic Orbitals

You may be wondering why the **order of orbital energy levels is not entirely determined by the total number of nodes.** After all, we have established that the energy of a wave depends on its frequency—and by extension, its number of nodes. As shown in Figure 15, the 2s and 2p orbitals illustrate this pattern well because both contain the same number of nodes (1), yet the 2s orbital sits at a lower energy than the 2p.

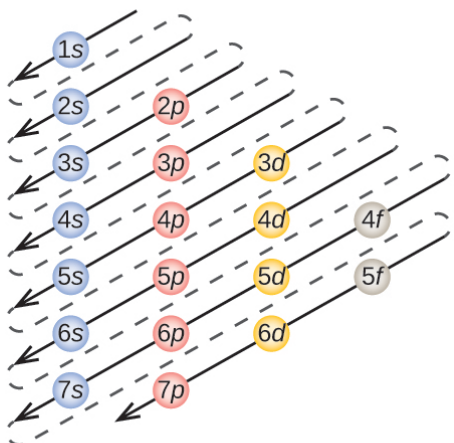

Figure 15: Electron orbital energy level ordering schematic

To understand why, we first examine **hydrogenic atom orbitals**—those of atoms stripped of all but one e^-, such as H, He^+, and Li^{2+}. In these simplified systems, the energy of the electron depends *only* on the total number of nodes, or equivalently, on the principal quantum number n, regardless of the angular quantum number l. This is fully consistent with our earlier observation that more nodes means higher frequency, which means higher energy. The energy of a hydrogenic orbital is given quantitatively by:

$$E = -\frac{hcRZ^2}{n^2}$$

where h is Planck's constant (6.626×10^{-34} J s), c is the speed of light (2.998×10^8 m s^{-1}), R is the Rydberg constant (1.097×10^7 m^{-1}), Z is the atomic number, and n is the principal quantum number. The negative sign reflects the fact that the electron sits in an energy well relative to a free electron—consistent with our discussion of potential energy and bond energies in Core Concept 1.3.

Two trends follow directly from this expression. First, **energy decreases as Z increases**, reflecting the stronger attractive pull of a more positively charged nucleus on the single e^-. Second, energy increases with n, consistent with the wave-like nature of the electron and the role of nodes in determining frequency. Figure 16 below illustrates these energy levels across different principal quantum numbers for a hydrogenic atom.

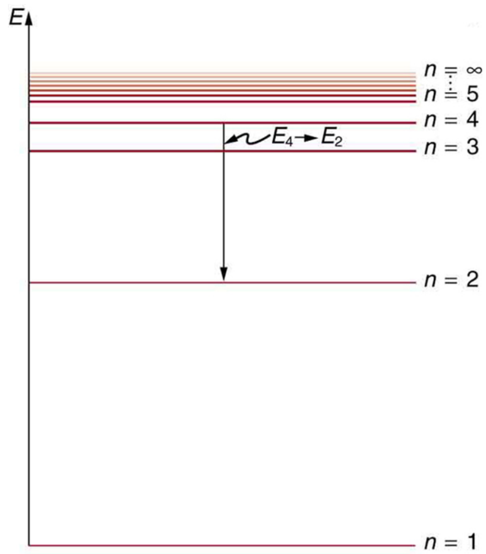

Figure 16: Hydrogen-like atom orbital energies across different principal quantum numbers

The energy change when an e^- transitions between orbitals is given by the following equation

$$\Delta E = -hcRZ^2\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

With this foundation in place, we can now return to the original question. In a hydrogenic atom, orbital energy depends solely on *n*. However, when additional electrons are introduced, **e^- e^- repulsion** disturbs this clean dependence. These repulsion interactions alter the **radial** character of orbitals—while the **angular** character and total node count **remain unchanged** from the hydrogenic case—causing orbitals with the same *n* but different *l* to diverge in energy. The result is that orbital energy in multi-electron atoms depends on *both n* and *l*, producing the ordering we observe in practice.

Core Concept 2.8: Aufbau Principle

According to the **Aufbau principle**, e^- fill orbitals in order from **lowest to highest energy** as an atom gains more e^-s. The diagonal arrow schematic in Figure 15 is a useful mnemonic for remembering this filling order across subshells.

Core Concept 2.9: Simplifying Electron Configuration Notation

The Aufbau principle is best understood through electron configuration diagrams, where orbitals are filled sequentially from lowest to highest energy. Figure 17 illustrates this clearly for Li, Ne, and Al.

It is also convenient to use **noble gas notation** to simplify electron configurations. By placing a noble gas symbol in brackets—for example, [Ne] as shown in the Al electron configuration—you represent all core electrons corresponding to that noble gas's stable configuration, allowing you to focus exclusively on the valence electrons responsible for chemical bonding.

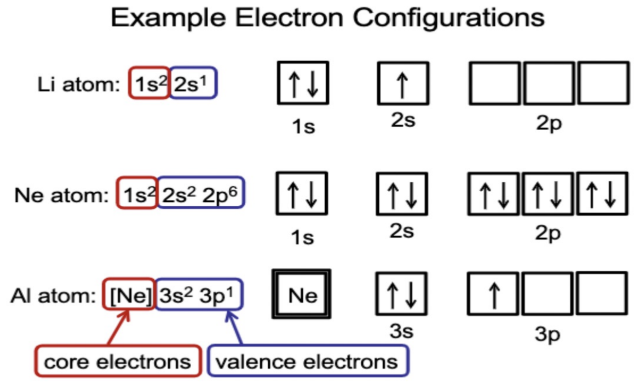

Figure 17: Electron configuration notation for Li, Ne, and Al

Core Concept 2.10: Hund's Rule

Hund's Rule states that within the highest energy subshell, e^-s occupy orbitals singly and with the same spin before any pairing occurs, maximizing the number of unpaired e^-s.

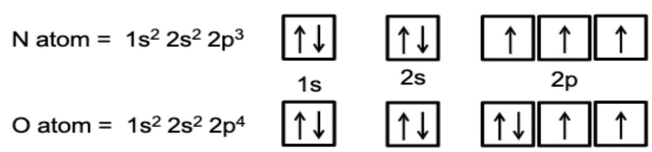

Figure 18: Hund's rule electron configurations

The underlying reason stems from the e^- e^- repulsion interactions briefly introduced earlier, the full treatment of which is beyond the scope of these notes.

Core Concept 2.11: Ionization Energy Analysis

To ground this in real numbers, consider the ionization energies of Mg and B, with respective electron configurations $[Ne]\,3s^2$ and $[He]\,2s^2\,2p^1$.

$E_{\text{ionization}}$ for Mg	
1st	7.6 eV
2nd	15.0 eV
3rd	80.1 eV

Table 2: Ionization energies of Mg

The sharp jump at the 3rd ionization energy reflects the transition from the 3s valence orbital to the 2p core orbital, which sits considerably deeper in the energy well. The same pattern emerges for B.

$E_{\text{ionization}}$ for B	
1st	8.3 eV
2nd	25.2 eV
3rd	37.9 eV
4th	259.4 eV

Table 3: Ionization energies of B

Here, the jump from the 1st to the 2nd ionization energy marks the transition from the 2p to the 2s orbital, and the dramatic jump from the 3rd to the 4th marks the transition from the 2s to the 1s orbital. *Core electrons are held so tightly that they almost never participate in chemistry.*

Core Concept 2.12: Periodic Table Organization of Valance Orbital Configurations

The structural layout of the periodic table is itself a visual map of valence electron orbital configurations, as shown in Figure 19.

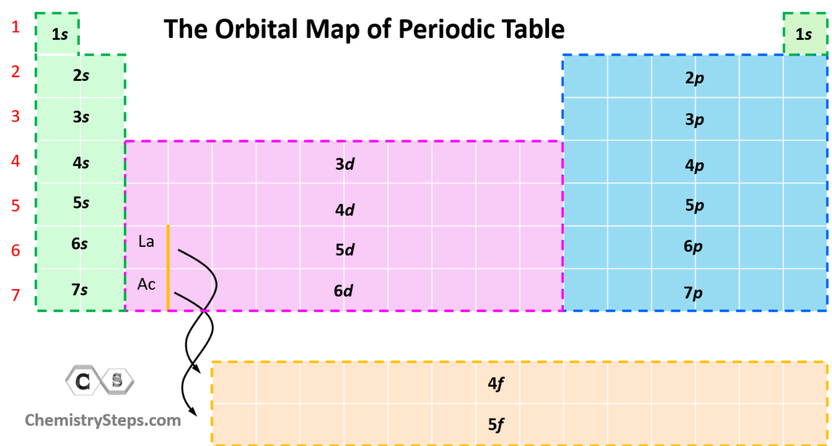

Figure 19: Periodic table of elements organization of valence electron orbitals

Elements in the same column share the same valence e^- configuration, which underlies their characteristically similar chemical behavior.

2.3 Electron Energy Emissions & Absorptions

Core Concept 2.13: Electron Energy Emissions & Absorptions

Returning to the electron energies of hydrogenic ion orbitals, the magnitude of the potential energy well of an e^- is given by the Rydberg formula.

$$E = -\frac{hcRZ^2}{n^2}$$

As a side note, multiplying by Avogadro's number gives the same quantity per mole of electrons.

$$E = -\frac{hcRN_A Z^2}{n^2}$$

Because electron bonding follows the same energetic logic as any bond, energy is absorbed when an e^- is promoted to a higher shell (*greater n*) and emitted when it relaxes to a lower shell (*lower n*).

$$\Delta E_{electron} > 0 \rightarrow \text{absorption} \qquad \Delta E_{electron} < 0 \rightarrow \text{emission}$$

Since the energy gaps between adjacent shells narrow at higher n, electrons in higher principal quantum number orbitals are **more susceptible to thermal excitation**—this is especially relevant for elements in the lower rows of the periodic table.

Flames provide a practical illustration of this phenomenon. At flame temperatures, intramolecular bonds are broken and **electrons are promoted to higher orbitals**. As they relax back down, the energy difference is **released as photons**. Because the Rydberg formula ties electron energy differences directly to photon wavelength, different compounds produce distinct emission wavelengths—and when those wavelengths fall within the **visible spectrum**, they appear as characteristic colors.

Core Concept 2.14: Crystal Field Theory

The emission of photons within the visible spectrum is particularly well associated with **transition metals**, whose valence electrons occupy d orbitals. To understand why, we must first introduce Crystal Field Theory.

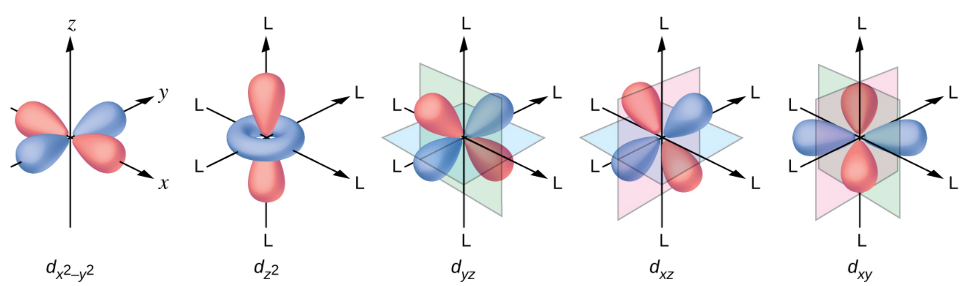

Figure 20: d orbital shapes

Crystal Field Theory is a model that explains the splitting of d orbital energy levels in **coordination compounds**—structures consisting of a central transition metal ion surrounded by bound molecules or ions called ligands, as shown in Figure 21.

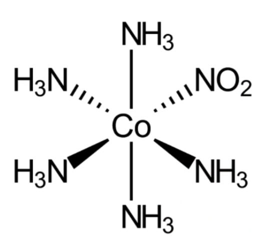

Figure 21: Coordination compound example

In the above mole, NH_3 and NO_2 act as ligands donating e^- density into the d orbitals of Co. This incoming e^- density repels the existing e^- density on the metal ion—the same e^- e^- repulsion discussed earlier in Core Concept 2.7—and just as that repulsion caused orbital energy levels to depend on both n and l rather than n alone, here it **unevenly distributes energy across the five d orbitals**, pushing some to higher energy levels than others.

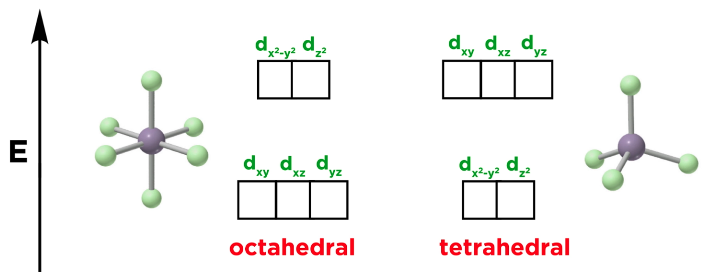

Figure 22: Crystal Field Theory d orbital energy level splitting

Crucially, the magnitude of this d orbital energy splitting in transition metals generally falls **within the energy range corresponding to visible light wavelengths**. This is not the case for non-transition metals, where electronic transitions span distinct orbital shells and subshells, producing energy differences far too large to fall within the visible spectrum and corresponding instead to much shorter wavelengths.

Underlying all of these interactions is the concept of **effective nuclear charge** (Z_{eff}). It also governs the shielding effect, whereby inner electrons occupy the space between the nucleus and outer electrons, repelling the outer ones and reducing the net nuclear attraction they experience—contributing directly to the skewing of orbital energy levels shown in Figure 22.

It should be noted that the treatment above is a simplified overview of Crystal Field Theory. A more complete picture accounts for the nature of the specific ligands present and the geometry of the coordination compound, both of which influence the magnitude and ordering of d orbital splitting.

3 Orbital Bonding

3.1 Sigma/Pi & Bonding/Antibonding Orbitals

Core Concept 3.1: Orbital Wave Overlap

Bonding is favorable because it results in a lower energy state, as discussed in Core Concept 1.3. Now that we have established that orbitals represent electron density, we can explain the physical origin of this energy drop from both an electrostatic and a wave mechanical perspective.

(a) From an **electrostatic standpoint**, when two atoms bond, electron density accumulates directly between two positive nuclei. Rather than being attracted to a single nucleus, each e^- is now simultaneously attracted to two — this bridge of **negative charge pulling on both positive centers lowers the overall PE of the system**.

(b) From a **wave mechanical standpoint**, bonding can be understood through the **de Broglie relationship**, which is represented by the following equation.

$$\lambda = \frac{h}{mv}$$

This equation links the wavelength of an e^- to its momentum and thus its KE. When two atomic wave functions overlap during bond formation, they undergo interference—**constructive interference** in the bonding region increases the combined wavelength of the resulting wave function, and since a longer wavelength corresponds to lower frequency and slower effective motion, the KE of the e^- is reduced.

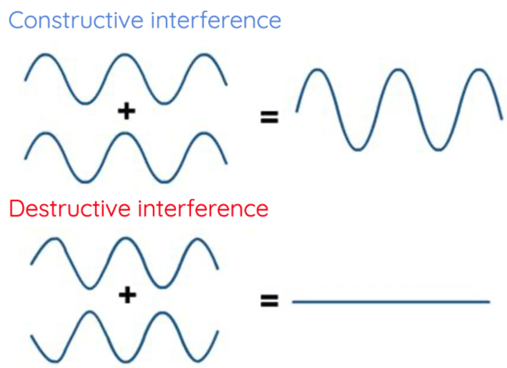

Figure 23: Constructive and destructive interference as a result of wave superposition

This is equivalent to enlarging the "box" that the e^- standing waves inhabit—a larger confining region permits a longer standing wave, recalling that the fundamental standing wave wavelength is twice the length of its confinement ($\lambda = 2L$).

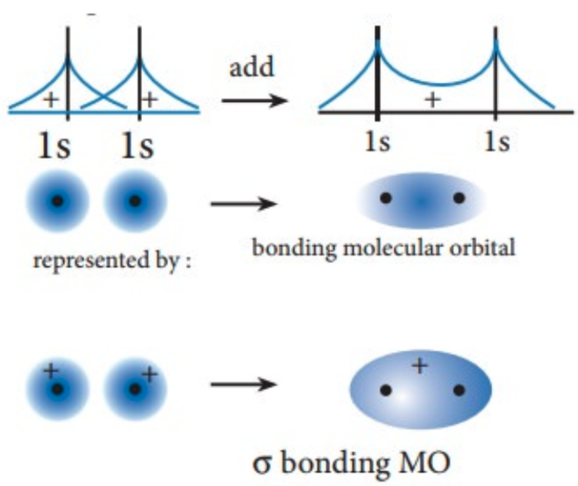

Figure 24: Molecular orbital formation wave overlap

Together, the reduction in both PE and KE is sufficient to overcome the nuclear-nuclear repulsion at the equilibrium bond length, making bond formation energetically favorable overall.

Core Concept 3.2: σ and π Bonds

Electron orbital overlap gives rise to two types of bonds, distinguished by the axis along which the overlap occurs: σ bonds (*head-on overlap*) and π bonds (*side-to-side overlap*).

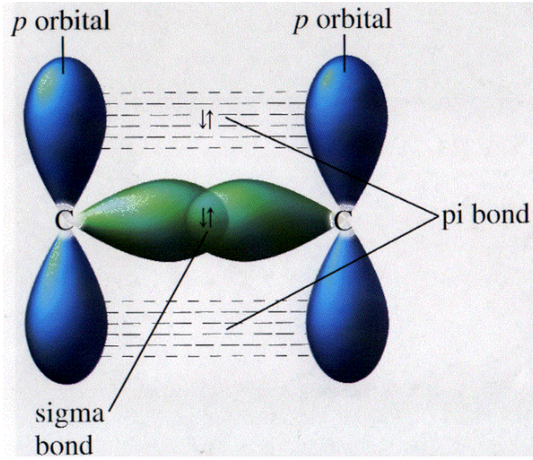

Figure 25: σ and π bonding interactions

It is important to note that σ and π do not refer to the s and p orbital types—they refer exclusively to **the geometry of the overlap**. A σ bond, for instance, can form between any combination of orbital types (s-s, s-p, p-p, etc.). For the scope of these notes, we will restrict our discussion to s and p orbital interactions, as d and f orbital bonding lies beyond the current scope.

(a) A **σ bond** forms when two orbitals overlap directly along the internuclear axis—that is, head-on from the center of one atom to the center of the other. The various orbital combinations that can produce a σ bond are shown in Figure 26.

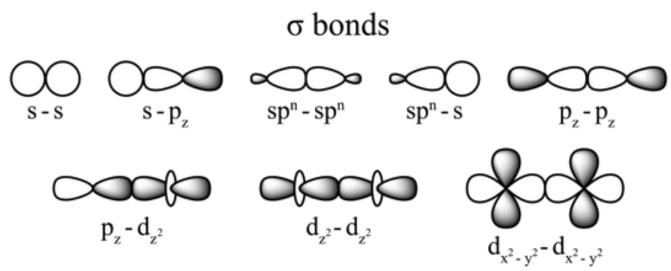

Figure 26: σ bond formation from various orbital overlap combinations

(b) A **π bond**, by contrast, forms when two orbitals overlap perpendicular to the internuclear axis. While various orbital combinations can produce a π bond, the most common involves two p orbitals overlapping side-to-side, as shown in Figure 27.

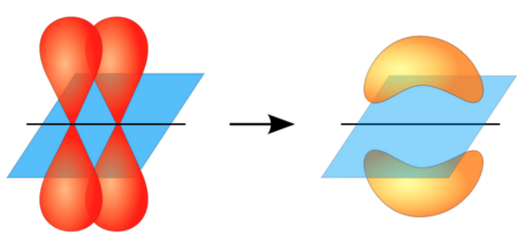

Figure 27: π bond formation from p orbital overlap

With this framework in place, single bonds consist of one σ interaction, while double and triple bonds layer one or two π bonds on top of the initial σ framework. This will become clearer in the context of Subsection 3.2.

Core Concept 3.3: Antibonding Orbitals & Quantum Superposition

Every orbital overlap that produces a **bonding orbital** simultaneously produces a corresponding **anti-bonding orbital**, denoted with an asterisk (σ^* or π^*). This is a consequence of **quantum superposition**—the overlapping wave functions do not simply choose between constructive and destructive interference; rather, the mathematical space defined by the two overlapping atoms permits both modes of vibration to exist **simultaneously** as distinct solutions.

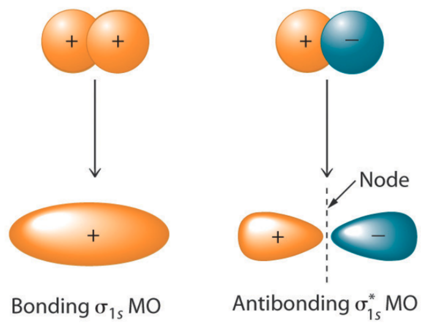

Figure 28: σ bonding and antibonding orbitals

This is formally captured by the **Linear Combination of Atomic Orbitals (LCAO)** theory, which requires that the **total number of orbitals be conserved**, meaning that if you begin with two atomic orbitals, you must end with two molecular orbitals—one bonding and one anti-bonding. The anti-bonding orbital carries an additional node relative to its bonding counterpart, placing it at a higher energy level. As a result, electrons fill the bonding orbital preferentially before occupying the anti-bonding orbital, consistent with the **Aufbau principle** discussed in Core Concept 2.8.

The σ bonding and anti-bonding orbitals are shown in Figure 28, and their π counterparts in Figure 29.

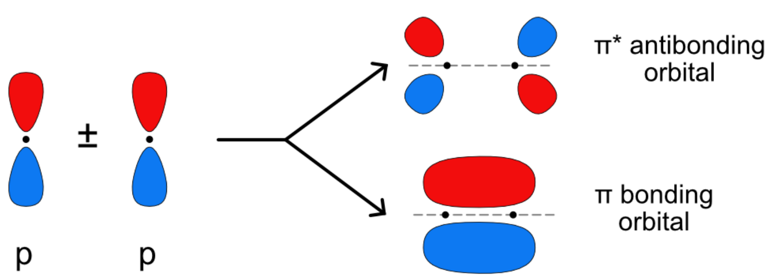

Figure 29: π bonding and antibonding orbitals

3.2 Diatomic Orbital Bonding

Core Concept 3.4: Homogenous Diatomic Molecular Orbital Bonding (*Single Bonds*)

When two atoms bond, it is their **valence orbitals**—the highest energy occupied orbitals containing unpaired electrons—that drive the interaction. Consider the simplest case of orbital overlap: the formation of H_2.

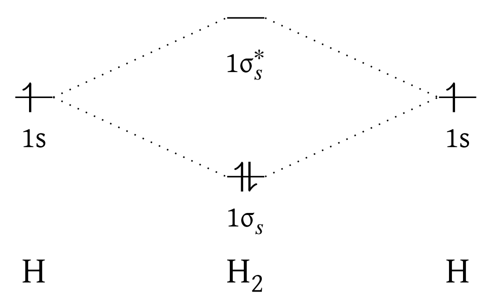

Figure 30: Molecular orbital diagram for H_2

The highest occupied atomic orbital in each hydrogen atom is the 1s orbital. When these overlap, they combine to form a low-energy bonding orbital σ and a high-energy antibonding orbital σ*. The two electrons from the two hydrogen atoms occupy the σ orbital, leaving σ* empty.

This leads naturally to the concept of **bond order**, a numerical measure of the stability and strength of a chemical bond representing the net number of bonds between two atoms.

$$\text{Bond Order} = \frac{\text{Number of bonding e}^- - \text{Number of antibonding e}^-}{2}$$

For H_2, this gives a bond order of 1, consistent with a **single bond**. A bond order of 0 indicates that no stable bond forms, as the bonding and antibonding contributions cancel. This is illustrated by He_2.

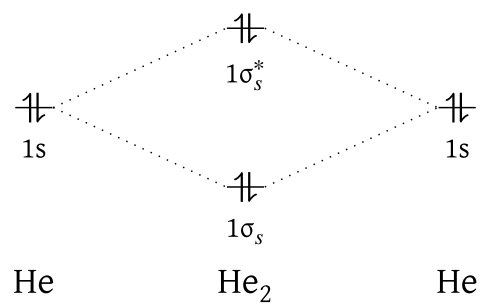

Figure 31: Molecular orbital diagram for He_2

In He_2, electrons occupy both the bonding and antibonding orbitals, yielding a bond order of 0. Since the two contributions cancel, no net bond forms—consistent with the fact that He_2 does not exist naturally.

Core Concept 3.5: Homogenous Diatomic Molecular Orbital Bonding (*Double Bonds*)

Consider now the orbital overlap in O_2, which gives rise to a **double bond**.

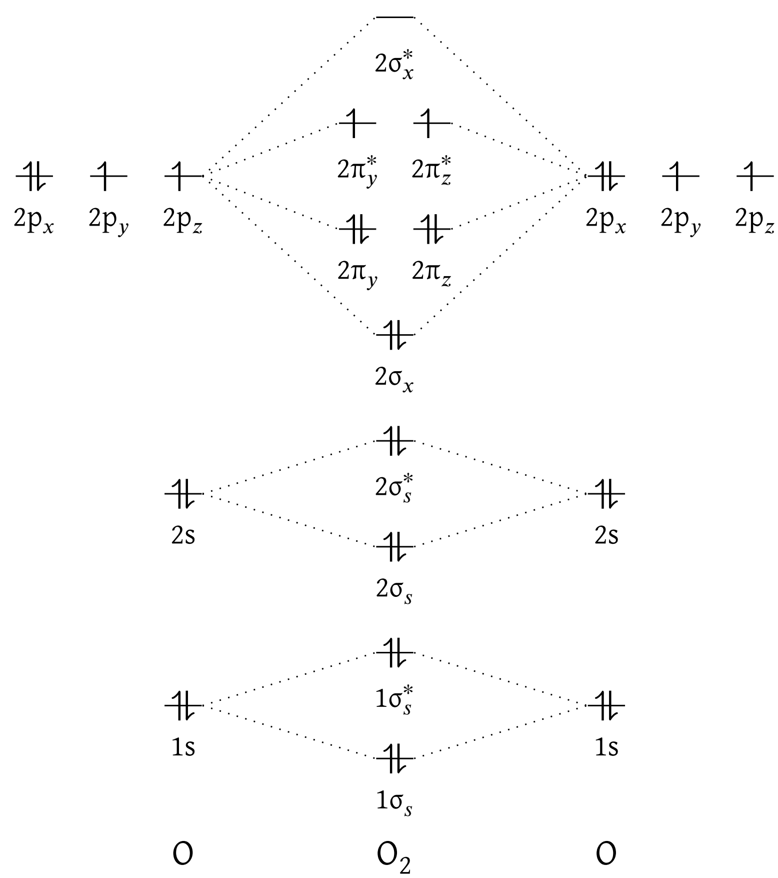

Figure 32: Molecular orbital diagram for O_2

The lower-energy 1s and 2s orbitals each produce fully occupied bonding and antibonding pairs that cancel out, contributing nothing to the net bond—consistent with the fact that these are not the valence orbitals. It is the 2p orbitals that drive bonding. Of the three spatial orientations of the 2p orbitals, those aligned along the bonding axis overlap to form σ and σ* orbitals, while those perpendicular to it overlap to form π and π* orbitals. Filling these according to the Aufbau principle, all three bonding orbitals are fully occupied and one electron is placed in each of the two π* orbitals, giving a bond order of 2.

Core Concept 3.6: Homogenous Diatomic Molecular Orbital Bonding (*Triple Bonds*)

Consider finally the molecular orbital diagram of N_2, which gives rise to a **triple bond**. The orbital structure follows the same logic as O_2 (*Core Concept 3.5*), with the key distinction that no electrons occupy any antibonding orbitals. All six bonding electrons are therefore uncontested, yielding a bond order of 3.

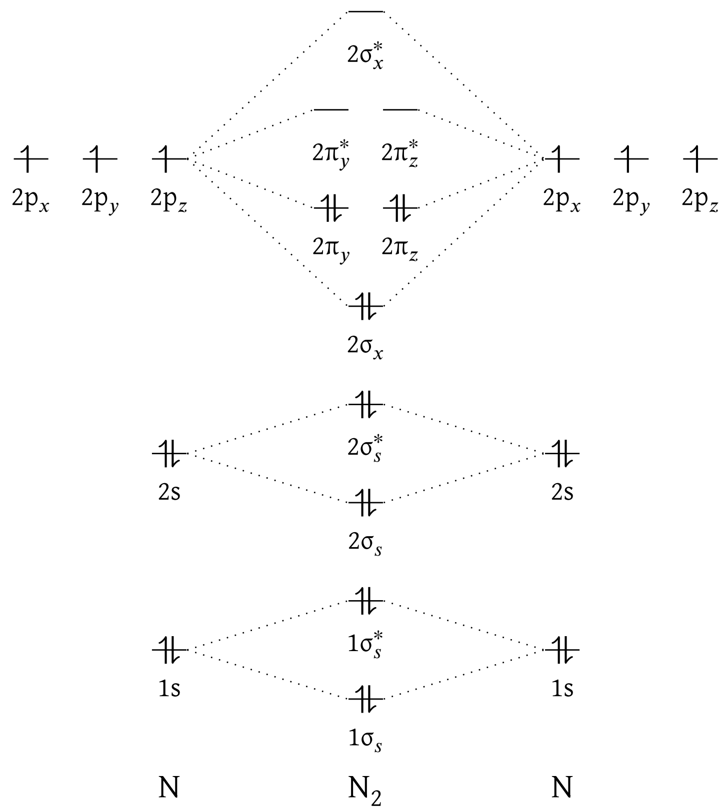

Figure 33: Molecular orbital diagram for N_2

Core Concept 3.7: Heterogenous Diatomic Molecular Orbital Bonding

We now turn to **heterogeneous** (*different elements*) interactions, beginning with HF.

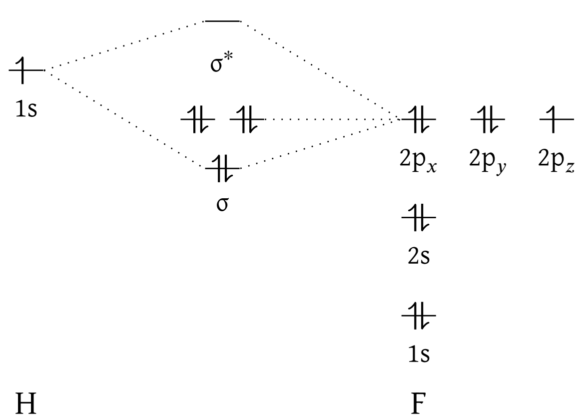

Figure 34: Molecular orbital diagram for HF

Here, the only possible interaction is a σ bond formed between the 1s orbital of H and the 2p orbital of F aligned along the bonding axis—a straightforward heterogeneous interaction. Consider now a slightly more complex case, ClF, which introduces several lower-energy core orbitals.

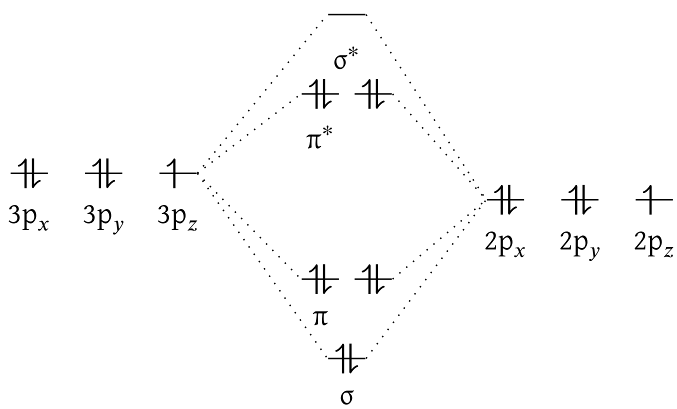

Figure 35: Molecular orbital diagram for ClF; *non-valence orbitals exlcuded*

Even here, **bonding interactions occur only between the incompletely occupied valence orbitals**—the 2p of F and the 3p of Cl. The lower-energy core orbitals do form bonding and antibonding pairs, but since these contributions cancel to a net bond order of zero, they have no effect on the valence electronic structure.

More generally, while core orbitals do overlap to some infinitesimal degree whenever two atoms are in close proximity, this interaction is negligibly small and is therefore treated as nonexistent.

Core Concept 3.8: S-P Mixing (*Periodic Table 2nd Row*)

Along the second-row diatomic molecules, a phenomenon known as **s-p mixing** produces an anomaly in the relative energy ordering of the σ_{2p} and π_{2p} orbitals. It arises from the interplay between e^- e^- repulsion and e^- nuclear attraction at this range of nuclear charge.

In lighter diatomic molecules such as B_2, C_2, and N_2, the lower effective nuclear charge narrows the energy gap between the 2s and 2p orbitals sufficiently for significant mixing to occur. This pushes the σ_{2p} orbital higher in energy than the π_{2p}, inverting the usual ordering. The molecular orbital diagram of N_2 shown earlier (Figure 33) does not account for this; a corrected version is shown below in Figure 36.

In heavier molecules such as O_2 and F_2, the greater effective nuclear charge widens the 2s–2p energy gap enough to suppress mixing, leaving π_{2p} higher in energy than σ_{2p} as expected. S-p mixing carries two notable practical consequences.

- **The exceptional stability of N_2**: Strong s-p mixing in nitrogen lowers the energy of the σ_{2s} bonding orbital and raises that of σ_{2p}, stabilising the molecule significantly. This is partially why N_2 is so chemically inert and constitutes the majority of Earth's atmosphere—without this effect, nitrogen may be far more reactive, with profound consequences for life.
- **Magnetism prediction**: S-p mixing also bears on the magnetic properties of B_2, which is discussed in Core Concept 3.9.

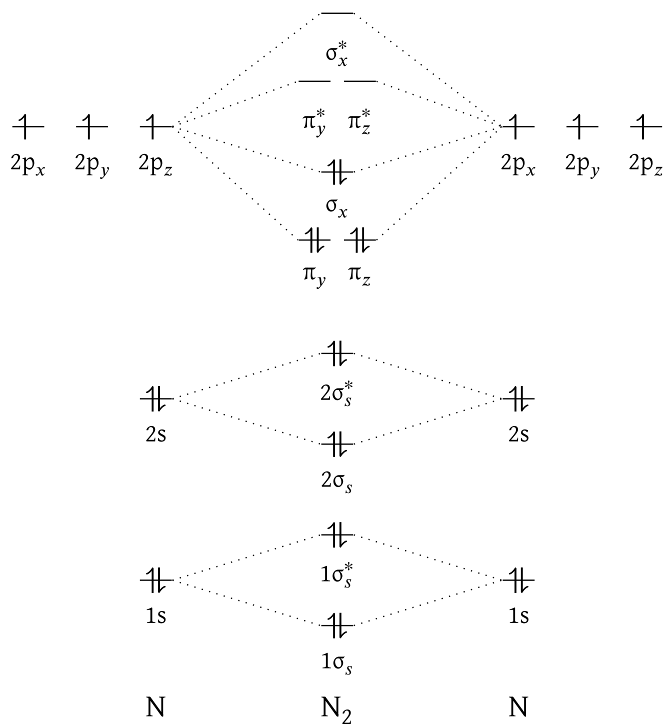

Figure 36: Corrected molecular orbital diagram for N_2

Core Concept 3.9: Diamagnetism & Paramagnetism

A molecule is **diamagnetic** if it is not attracted to a magnetic field and **paramagnetic** if it is, a distinction determined entirely by the presence or absence of unpaired electrons in its molecular orbitals.

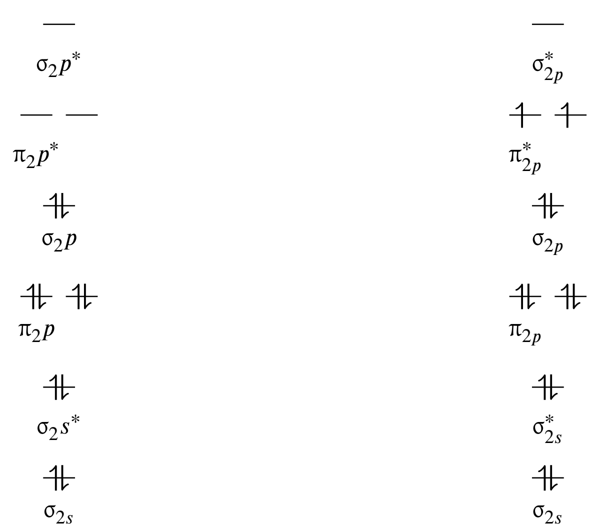

Figure 37: N_2 (*left*) being diamagnetic (*not attracted to magnetic field*) due to the absence of unpaired e^-s; O_2 (*right*) being paramagnetic (*attracted to magnetic field*) due to the presence of unpaired e^-s at the π^*_{2p} molecular orbitals

To understand why unpaired electrons produce magnetism, we turn to **spin**—the intrinsic angular momentum of an electron, briefly introduced through the spin quantum number m_s in Core Concept 2.5. Each electron generates a small magnetic field oriented along its spin direction. When two electrons of opposite spin occupy the same orbital, their magnetic fields cancel, yielding a net field of zero. An **unpaired electron**, however, has no opposing spin to cancel it, so its magnetic moment persists and the molecule is **attracted to external magnetic fields**.

The molecular orbital diagram of B_2 provides a striking illustration of this.

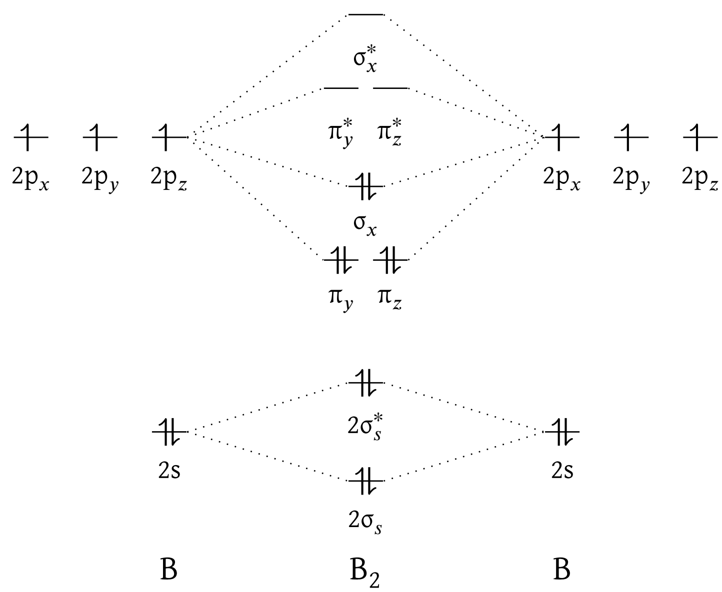

Figure 38: Molecular orbital diagram for B_2

The paramagnetism of B_2 is a **direct consequence of s-p mixing**. Without it, the two valence electrons would pair up in the σ_{2p} orbital with opposite spins, producing a diamagnetic molecule. Because s-p mixing lowers the π_{2p} orbitals below σ_{2p}, however, these electrons instead occupy the two degenerate π_{2p} orbitals one each with the same spin—giving rise to paramagnetism.

Core Concept 3.10: Bonding & Antibonding Orbital Energy Split

The size of the energy split between bonding and antibonding molecular orbitals is determined by the degree of overlap between the two interacting orbitals—greater overlap produces a larger split. Several factors govern the strength of this overlap.

(a) σ bonds overlap more strongly than π bonds because they interact directly along the bonding axis, producing a larger energy split.

(b) Smaller orbitals (*e.g.* $n = 2$) overlap more intensely than larger ones (*e.g.* $n = 5$) due to greater electron density, closer internuclear proximity, and shorter inter-orbital distances. This explains why the HOMO–LUMO (*Core Concept 3.16*) excitation energy among the homonuclear diatomic halogens decreases down the periodic table.

$$E_{F_2} > E_{Cl_2} > E_{Br_2} > E_{I_2}$$

(c) A smaller energy difference between the interacting atomic orbitals leads to greater overlap, while a larger difference—arising from **electronegativity differences**—skews electron density toward one atom rather than delocalising it uniformly. As a result, homonuclear bonds tend to be more stable than heteronuclear bonds, and their ionisation energies (IE) are generally higher. For instance, despite CO and N_2 having virtually identical molecular orbital diagrams, their IEs are approximately 14 eV and 15.6 eV respectively. Similarly, despite O_2 and NO sharing the same HOMO (π^*_{2p}), their IEs differ markedly at approximately 12 eV and 9.2 eV respectively.

3.3 Orbital Hybridization

Core Concept 3.11: Orbital Hydridization

Orbital hybridization is the mixing of atomic orbitals to form a new set of equivalent orbitals. The number of orbitals is conserved and they remain orthogonal to one another. It serves as the correction factor for the mismatch between how atomic orbitals appear in isolation and how they behave when participating in multiple bonds simultaneously.

Core Concept 3.12: ***sp*** **Hybridization**

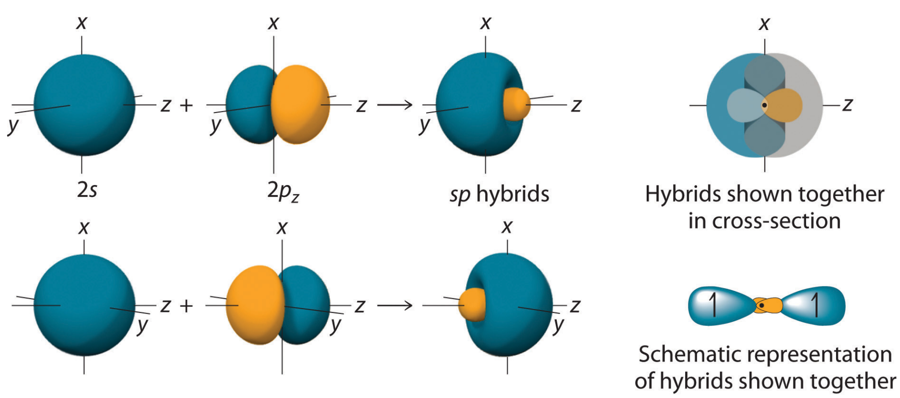

Figure 39: sp hybridization showing orbital mixing

The simplest case of hybridization is *sp*, formed by mixing **one s and one p orbital** to produce two equivalent orbitals arranged at 180° to each other. These orbitals accommodate two σ bonds and/or lone pairs in a linear geometry. A representative example is HCN, with the MO configuration $(1s)^4(\sigma_{CH})^2(\sigma_{CN})^2(\pi_{CN})^4(n_N)^2$

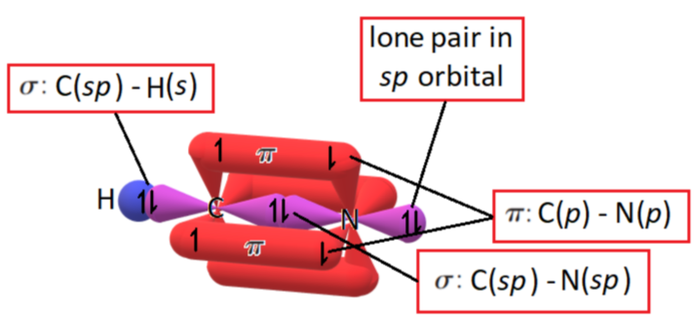

Figure 40: HCN molecule showing sp hybridization

Core Concept 3.13: sp^2 Hybridization

sp^2 hybridization results from mixing **one s and two p orbitals**, producing three equivalent orbitals pointing toward the vertices of an equilateral triangle at 120° to each other. These orbitals accommodate three σ bonds and/or lone pairs, while the remaining unhybridized p orbital is available for π bonding. A representative example is ethylene C_2H_4, with the MO configuration $(1s)^4(\sigma_{CH})^8(\sigma_{CC})^2(\pi_{CC})^2$.

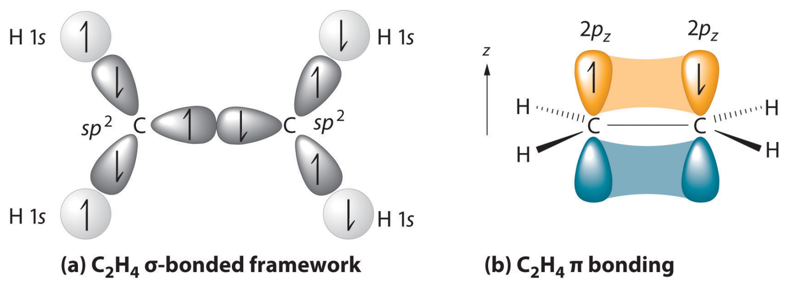

Figure 41: Ethylene molecule showing sp2 hybridization

Core Concept 3.14: sp^3 Hybridization

sp^3 hybridization results from mixing **one s and three p orbitals**, producing four equivalent orbitals arranged tetrahedrally at 109.5° to each other, accommodating four σ bonds and/or lone pairs. A representative example is CH_3F, with the MO configuration $(1s)^4(\sigma_{CH})^6(\sigma_{CF})^2(n_F)^6$.

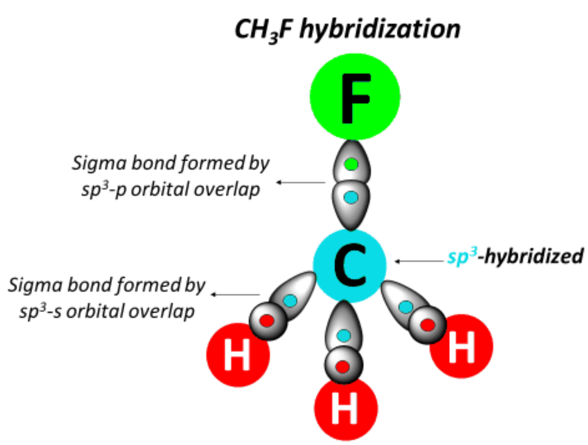

Figure 42: CH3F molecule orbital diagram

Orbital hybridization occurs because bonding interactions proceed in whatever manner **minimizes the system's potential energy**. Hybrid orbitals are asymmetric—featuring one large lobe and one small lobe—allowing them to extend further into space than standard s or p orbitals. This **greater spatial extension produces more overlap** with neighboring orbitals, increasing the energy gap between bonding and antibonding MOs and thereby lowering the overall potential energy of the system.

3.4 Delocalization, Resonance, and Light

Core Concept 3.15: Resonance & Delocalization

Consider the Lewis structure of benzene (C_6H_6), which can be drawn in two equivalent ways.

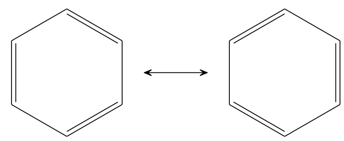

Figure 43: Benzene resonance structures

These interchangeable representations are called **resonance structures**. From a molecular orbital perspective, benzene consists of a plane of sp^2 orbitals with six 2p orbitals oriented perpendicular to the following plane.

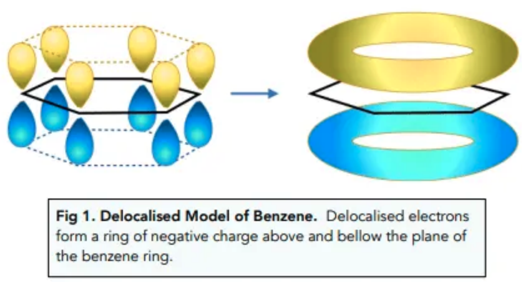

Figure 44: Benzene with delocalized π system

The localized π bonds implied by the resonance structures in Figure 43 are in fact a simplification—in reality, the e^-s within the parallel 2p orbitals **delocalize** continuously across the entire ring (*Figure 44*) rather than residing between any two specific atoms. The following figure further illustrates this occurance.

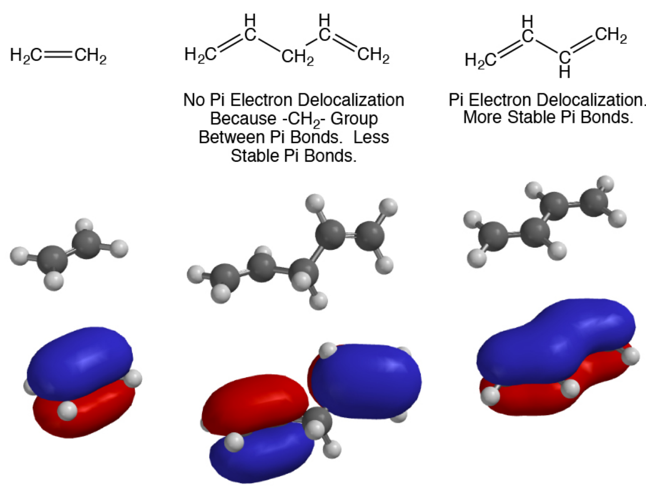

Figure 45: Examples of resonance and delocalization

Core Concept 3.16: Light & HOMO LUMO

The acronyms HOMO and LUMO stand for the **highest occupied molecular orbital** and **lowest unoccupied molecular orbital**, respectively, and the **HOMO-LUMO gap** is the energy difference between them. Consider the molecular orbital diagram for O_2.

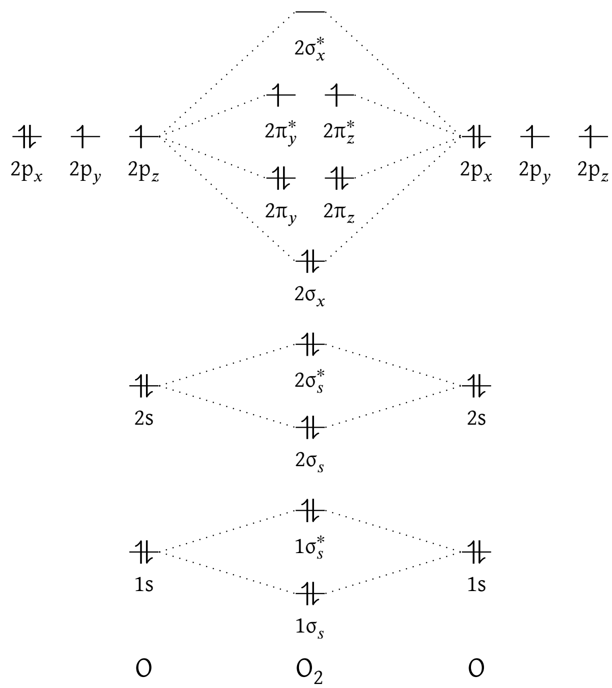

Figure 46: Molecular orbital diagram for O_2

Here the HOMO is π_{2p}^* and the LUMO is σ_{2p}^*. The HOMO-LUMO gap represents the **energy required to excite an electron from the HOMO to the LUMO**, and its magnitude determines which wavelength of light the molecule absorbs. This is illustrated clearly by comparing the MO diagrams of halogens (*Group 17*).

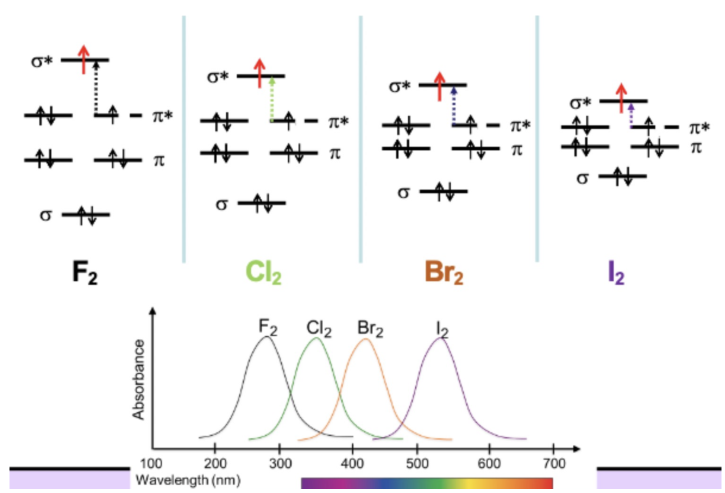

Figure 47: HOMO-LUMO orbitals in halogens

As the HOMO-LUMO gap widens, the excitation energy increases and the absorbed wavelength decreases. When white light—a mixture of all visible wavelengths—is incident on the molecule, what we perceive is the complement of the absorbed wavelengths. For example, I_2 absorbs in the green-yellow region, so we observe its complementary color, purple. The same logic applies to Cl_2 and Br_2. In the case of F_2, its HOMO-LUMO gap corresponds to UV wavelengths, meaning no visible light is absorbed and nearly all wavelengths are reflected—producing a pale, near-transparent appearance with a slight yellow tint from partial absorption of violet light.

Figure 48: Halogen gases in order: F2, Cl2, Br2, I2

Combining this with our understanding of delocalization (*Core Concept 3.15*), greater delocalization is associated with a **smaller** HOMO-LUMO gap. Although delocalization stabilizes the system through increased overlap, the energy intervals between successive MO levels in the conjugated π-system become smaller as more p orbitals are included—even as the total energy span from the lowest to highest MO increases. This is illustrated below by comparing the π-system MO diagrams of ethene (*two p orbitals*) and butadiene (*four p orbitals*).

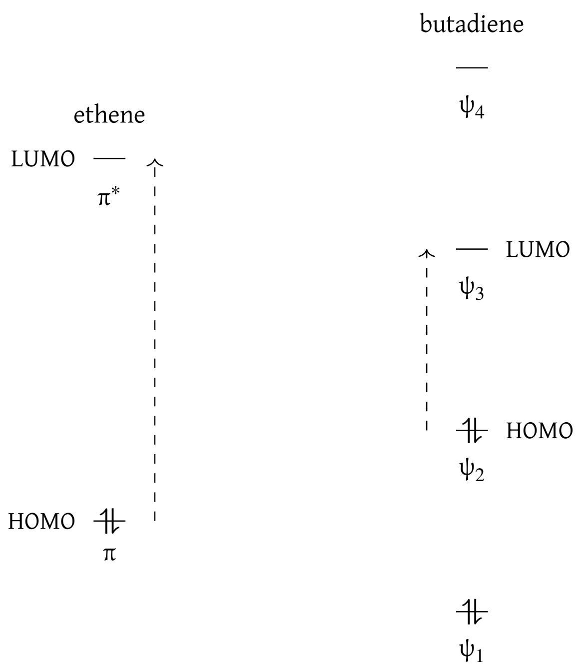

Figure 49: HOMO-LUMO delocalization in conjugated systems of ethene (H_2CCH_2) and butadiene ($H_2CHCHCH_2$)

Core Concept 3.17: Infrared Spectroscopy

Molecules do not only absorb light through electronic transitions—they also absorb in the **infrared spectrum**, where the energy is too low to excite e⁻s but sufficient to increase the **kinetic energy** of the molecule through vibrational motion: stretching, bending, and rotating of bonds.

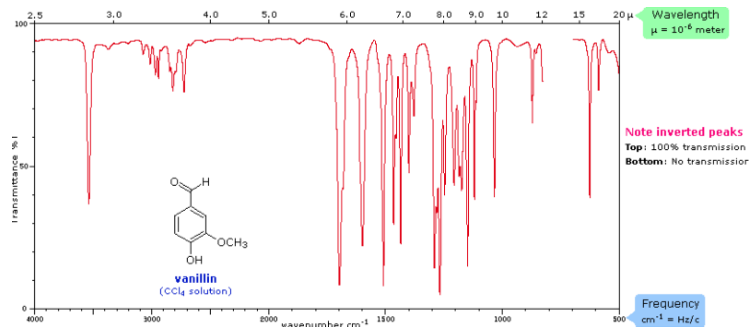

Figure 50: Infrared spectroscopy fundamentals

Figure 50 is an infrared spectrum for vanillin, where each peak corresponds to a specific wavelength of infrared light absorbed by the molecule to drive a particular vibrational mode—analogous to how electronic transitions produce discrete absorption peaks in the visible spectrum. When viewed across a broader wavelength range, these individual peaks merge into characteristic broad absorption bands.

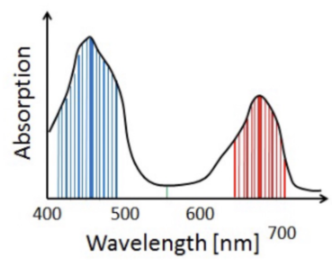

Figure 51: IR absorption bands for functional groups

Core Concept 3.18: Proton Nuclear Magnetic Resonance

Just as e⁻ possess spin, so do protons. In the absence of an external magnetic field their spins are randomly oriented, but when placed in one, each proton aligns either parallel (*lower energy*) or anti-parallel (*higher energy*) to the field. The energy gap between these two states grows with the strength of the applied field:

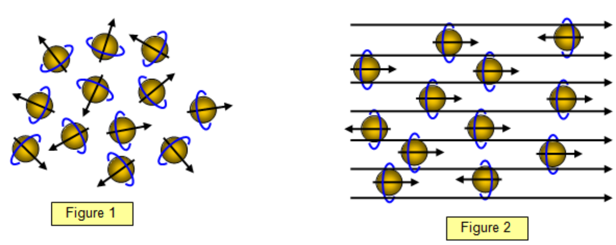

Figure 52: Proton nuclear magnetic resonance basics

By irradiating these protons with radiofrequency radiation matched to this energy gap, their spins can be induced to flip between states. The energy released as they relax back to the lower-energy orientation is precisely what **magnetic resonance imaging (MRI)** detects and uses to construct detailed maps of tissue in the body.